

FleetOps AI

Demo Walkthrough Guide

A companion guide to the demo video (recorded without narration), explaining what is shown on screen at each step

Track: AI + Construction Machinery → AI + Operations

This guide follows the same order as the recorded demo video. Each section below corresponds to one part of the video and explains what is being shown on screen.

1. Introduction & Overview

Hello everyone! Today, I will demonstrate our project, FleetOps AI, an intelligent operations system for construction equipment. Our goal is to help managers monitor equipment performance, identify potential problems, and make better maintenance decisions.

We use a sample dataset containing 30 equipment units and 90 days of operational data. First, on the Overview page, we can see the main fleet KPIs, including fleet size, utilization, operating hours, and fuel consumption. We can also compare equipment downtime and utilization across different projects.

2. Ask the AI Agent

Next, we have our AI Agent, which allows managers to ask questions about fleet performance using natural language. For example, I will ask the agent to identify the top three equipment units that need maintenance attention. As you can see, the system identifies the equipment, explains why each unit needs attention, and recommends an action. The results are presented in clear cards, making them easy to understand.

The system uses built-in analytics and machine learning, and we can also connect an optional language model for more advanced conversations.

DEMO QUESTION (shown being typed into the chat in the video)

"Rank the top 3 equipment units that need maintenance attention today. For each one, explain the reason and recommend the next action."

3. AI Maintenance Planner

Now, let's move to the AI Maintenance Planner. This feature creates a seven-day maintenance plan based on the equipment's Attention Score. Here, I can choose how many equipment units I want to include. For example, if I select five, the system automatically selects the five highest-priority units, classifies them by priority, assigns maintenance days, and estimates the required service hours.

This helps managers organize maintenance activities instead of selecting equipment manually.

4. What-If Maintenance Simulator

Our next feature is the What-If Maintenance Simulator. This allows managers to explore what might happen if they delay maintenance. For this demonstration, I will select equipment EQ-024 and delay its maintenance by nine days. The simulator estimates the impact on downtime and utilization.

In this scenario, additional downtime is estimated at 1.94 hours, and utilization decreases by 2.8 percentage points. The system also provides a recommended decision.

These results are scenario estimates, not guaranteed predictions, and they help managers compare options before making a decision.

5. Report

Finally, we have the Report section, which summarizes fleet performance and highlights important operational issues and recommended actions. It helps managers review the results in one place.

6. Conclusion

To conclude, FleetOps AI is more than just a dashboard. It combines data analytics, machine learning, maintenance planning, and what-if simulation to support operational decision-making. Our goal is simple: to turn fleet data into useful decisions.

Thank you for watching our demo!

This document accompanies the recorded demo video for the China International College Students' Innovation & Entrepreneurship Competition (Track: AI + Construction Machinery → AI + Operations).